RATAN TATA MAHARASHTRA STATE SKILLS UNIVERSITY

School of Science Engineering and Technology Skills Department of Engineering

Subject –DevOps and Automation Lab

Lab Assignment 6

Aim:

Build a Docker Image for a Sample Application and Run it as a Container

Learning Objective:

- Understand Docker image creation process
- Learn how to containerize an application
- Execute and manage Docker containers

Prerequisites:

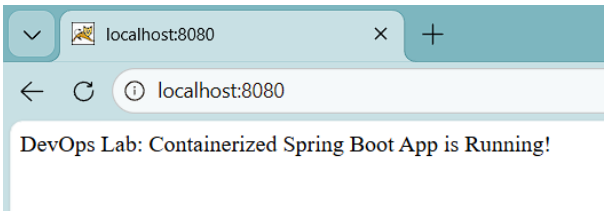
- Docker installed and configured (verify using docker --version)
- Basic knowledge of command line
- Any sample application (Python / Node.js / Java etc.)

```
❯❯ ~  16:02 docker --version
Docker version 29.2.0, build 0b9d198
```

Steps:

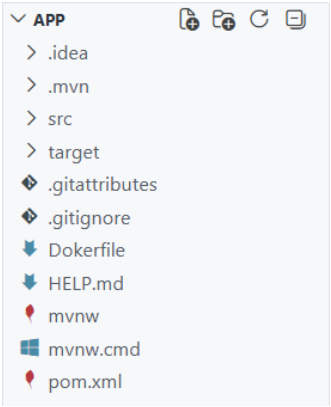
1. Understand the Application

- Select a simple sample application (e.g., Hello World web app)
- Identify application dependencies
- Test the application locally before containerizing



2. Create Project Structure

- Create a project folder
- Add application source code
- Add dependency files (e.g., requirements.txt, package.json)



3. Write a Dockerfile

- Create a file named Dockerfile
- Define base image (e.g., python, node, openjdk)
- Set working directory using WORKDIR
- Copy files using COPY
- Install dependencies using RUN
- Define default command using CMD

Dockerfile

Dockerfile > ...

```
FROM maven:3.9.9-eclipse-temurin-17 AS build

WORKDIR /app

# Copy Maven wrapper and project metadata first for better layer caching
COPY .mvn .mvn
COPY mvnw mvnw
COPY mvnw.cmd mvnw.cmd
COPY pom.xml pom.xml

RUN chmod +x mvnw && ./mvnw -DskipTests dependency:go-offline

COPY src src
RUN ./mvnw -DskipTests clean package

FROM eclipse-temurin:17-jre-alpine

WORKDIR /app
COPY --from=build /app/target/App-0.0.1-SNAPSHOT.jar app.jar

EXPOSE 8080
CMD ["java", "-jar", "/app/app.jar"]
```

4. Build Docker Image

- Use docker build command
- Assign a tag to the image
- Observe build process and layers
- Verify image creation using docker images

```
❯❯ App  16:30 docker build -t my-app:v1 .
[+] Building 354.7s (20/20) FINISHED                                                    docker:desktop-linux
 => [internal] load build definition from Dockerfile                                                   0.0s
 => => transferring dockerfile: 543B                                                                   0.0s
```

```
❯❯ App  16:36 docker images
                                                                                    i Info →   U  In Use
IMAGE                ID             DISK USAGE   CONTENT SIZE   EXTRA
my-app:v1            9b450cb20910       293MB         85.8MB
postgres:15-alpine   5fe8ca7fc662       392MB          110MB     U
```

5. Run Docker Container

- Use docker run command
- Map ports (if web app) using -p
- Run container in detached/interactive mode
- Assign container name

```
❯❯ App  16:38 docker run -d -p 8081:8080 --name my-running-app my-app:v1
696e0eb70c97d038cc32bef60838bc3cf453ebb49ccf5a24a1bf8d870a8b104e
```

6. Verify Container Execution

- Check running containers using docker ps
- Access application via browser (if applicable)
- View logs using docker logs

```
❯❯ App  16:40 docker ps
CONTAINER ID   IMAGE      COMMAND                  CREATED          STATUS          PORTS                                       NAMES
696e0eb70c97   my-app:v1  "/__cacert_entrypoin…"   55 seconds ago   Up 54 seconds   0.0.0.0:8081->8080/tcp, [::]:8081->8080/tcp   my-running-app
❯❯ App  16:41 docker logs 696e0eb70c97

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 :: Spring Boot ::                (v4.0.5)

2026-04-07T11:10:36.166Z  INFO 1 --- [App] [           main] com.example.app.AppApplication          : Starting AppApplication v0.0.1-SNAPSHOT using Java 1
7.0.18 with PID 1 (/app/app.jar started by root in /app)
```

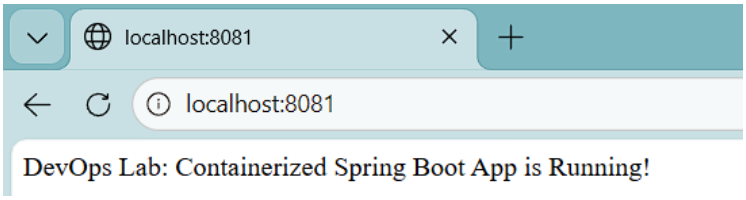

7. Manage Containers

- Stop container using docker stop
- Restart container
- Remove container using docker rm

```
❯❯ App  16:48 docker stop my-running-app
my-running-app
❯❯ App  16:48 docker start my-running-app
my-running-app
```

```
❯❯ App  16:53 docker stop my-running-app
my-running-app
❯❯ App  16:54 docker rm -f my-running-app
my-running-app
❯❯ App  16:54 docker ps -a
CONTAINER ID   IMAGE                COMMAND                  CREATED       STATUS                      PORTS     NAMES
c40ba146d067   postgres:15-alpine   "docker-entrypoint.s…"   2 weeks ago   Exited (137) 27 hours ago             dasf-postgres
```

8. Image and Container Inspection

- Inspect image using docker inspect
- Check container details
- Analyze image layers

```
›› App  16:48 docker inspect my-running-app
[
    {
        "Id": "696e0eb70c97d038cc32bef60838bc3cf453ebb49ccf5a24a1bf8d870a8b104e",
        "Created": "2026-04-07T11:10:34.771395359Z",
        "Path": "/__cacert_entrypoint.sh",
        "Args": [
            "java",
            "-jar",
            "/app/app.jar"
        ],
```

```
›› App  16:51 docker history my-app:v1
IMAGE          CREATED          CREATED BY                                      SIZE      COMMENT
9b450cb20910   16 minutes ago   CMD ["java" "-jar" "/app/app.jar"]              0B        buildkit.dockerfile.v0
<missing>      16 minutes ago   EXPOSE [8080/tcp]                               0B        buildkit.dockerfile.v0
<missing>      16 minutes ago   COPY /app/target/App-0.0.1-SNAPSHOT.jar app.…   19.8MB    buildkit.dockerfile.v0
<missing>      21 minutes ago   WORKDIR /app                                    8.19kB    buildkit.dockerfile.v0
<missing>      2 months ago     ENTRYPOINT ["/__cacert_entrypoint.sh"]          0B        buildkit.dockerfile.v0
<missing>      2 months ago     COPY --chmod=755 entrypoint.sh /__cacert_ent…   12.3kB    buildkit.dockerfile.v0
<missing>      2 months ago     RUN /bin/sh -c set -eux;     echo "Verifying…   12.3kB    buildkit.dockerfile.v0
<missing>      2 months ago     RUN /bin/sh -c set -eux;     ARCH="$(apk --p…   141MB     buildkit.dockerfile.v0
<missing>      2 months ago     ENV JAVA_VERSION=jdk-17.0.18+8                  0B        buildkit.dockerfile.v0
<missing>      2 months ago     RUN /bin/sh -c set -eux;     apk add --no-ca…   37.5MB    buildkit.dockerfile.v0
<missing>      2 months ago     ENV LANG=en_US.UTF-8 LANGUAGE=en_US:en LC_AL…   0B        buildkit.dockerfile.v0
<missing>      2 months ago     ENV PATH=/opt/java/openjdk/bin:/usr/local/sb…   0B        buildkit.dockerfile.v0
<missing>      2 months ago     ENV JAVA_HOME=/opt/java/openjdk                 0B        buildkit.dockerfile.v0
<missing>      2 months ago     CMD ["/bin/sh"]                                 0B        buildkit.dockerfile.v0
<missing>      2 months ago     ADD alpine-minirootfs-3.23.3-x86_64.tar.gz /…   9.11MB    buildkit.dockerfile.v0
```

9. Optimization

- Use smaller base images (e.g., Alpine)
- Reduce image size
- Use multi-stage builds

10. Clean Up

- Remove unused containers
- Remove unused images
- Use docker system prune

```
›› App  16:54 docker rmi my-app:v1
Untagged: my-app:v1
Deleted: sha256:9b450cb20910333d41b799977f603ae0a8d3a18f3a8880dfb90870a9628f0c02
›› App  16:55 docker images

IMAGE                ID             DISK USAGE   CONTENT SIZE   EXTRA
postgres:15-alpine   5fe8ca7fc662        392MB          110MB     U

›› App  16:55 docker system prune
WARNING! This will remove:
  - all stopped containers
  - all networks not used by at least one container
  - all dangling images
  - unused build cache

Are you sure you want to continue? [y/N] y
Deleted Containers:
c40ba146d067e629bd7ac9fa1a3aaa18a78076a88ca7eaa0064916b51757ae89
```